SILICON MOTION / CAD ENGINEERING TECHNICAL INTERVIEW

EDA / CAD Software Development & Physical Design Optimization

Logic Synthesis / Static Timing Analysis / Timing ECO

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EDA Tool Development / C++–Python–Tcl Automation / PPA Optimization

Education & Research Interests

Computer science and electronic physics backgrounds applied to IC design automation.

Education

National Taiwan University
M.S. Candidate, Data Science Master Degree Program
Sep. 2025 – Expected Jun. 2027

National Chengchi University
B.S. in Computer Science
Double Major in Electronic Physics
Sep. 2021 – Jun. 2025

Research Interests

EDA / CAD & VLSI Design Automation

Logic Synthesis and Technology Mapping
Static Timing Analysis / Timing Closure
Physical Design, Incremental ECO and PPA

Algorithm & Software Development

C++ / Python / Tcl Flow Automation
Digital IC / SoC Design and Verification

Technical interests: front-end synthesis, back-end timing optimization, and reproducible EDA flows.

Professional Software Engineering Experience

02 / WORK EXPERIENCE

Details index

Software development, simulation, and validation experience in R&D environments.

AIMobile Co., Ltd. (Inventec Group)

Apr. 2025 – Jul. 2025

Contract Software Engineer

Contributed to software development and system integration in an R&D environment.

Supported debugging and functional validation across software modules.

Neopower Technologies Co., Ltd.

Jun. 2023 – Mar. 2025

Contract Software Engineer

Developed a custom system simulator to reproduce operational and error scenarios.

Performed regression checks, log analysis, and customer-side A/B validation.

Selected Projects & Discussion Topics

Details index

Two projects connecting front-end technology mapping with back-end timing optimization.

01 Master's Research — Logic Synthesis

Multi-Output Technology Mapping in ABC `&nf` and GradMap

C++ EDA Tool Development / Boolean Matching / Area–Delay Optimization

02 ICCAD 2025 Problem C — CAD Methodology

STA-Driven Incremental Timing ECO Search

OpenROAD / OpenSTA / Python–Tcl Flow Automation / Heuristic PPA Search

Project 01: implemented MO discovery; area-aware selection is proposed.

Project 02: contest-inspired methodology under development.

Project Overview: NF-Direct MO Mapping

01 / RESEARCH SCOPE

Details index | SO baseline | GradMap integration

Discover FA/HA opportunities and globally pre-select MO candidates before GradMap.

Problem

Single-output cut retention does not explicitly optimize two-output FA/HA opportunities.

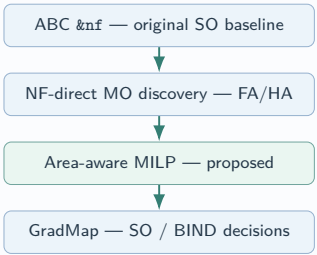
Our Extension

Add independent NF-direct MO discovery inside ABC `&nf`, without changing SO mapping.

Proposed Optimization

Solve an area-aware non-overlapping MO selection MILP before GradMap.

Research Pipeline



Input AIG + Liberty → selected.matches → mapped netlist

Why an Independent MO Discovery Path?

Details index SO baseline

SO-oriented cut pruning may remove cuts useful for joint FA/HA output matching.

Original NF SO Cut Ranking

Cuts are retained according to single-output objectives, not complementary FA/HA output pairs.

Independent NF-Direct Sidecar

Propagate additional MO cuts and CON/SN truth projections without modifying NF's SO path.

FIGURE SPACE

SO-pruned cut vs MO-retained cut

Two output roots with a shared legal 3-leaf input support.

MO discovery is not equivalent to scanning the final selected SO cover.

NF-Direct MO Candidate Discovery

04 / CORE ALGORITHM

Details index | Pair legality | SO witness

Pair compatible CON and SN projections over a common input support.

01 / MO Cut Propagation

Independent retention of FA/HA-relevant truth functions and supports.

02 / Projection Pairing

Match CON / SN roles using shared leaves, permutation and phase.

03 / Legality + Witness

Validate the pair and export v3 MOCAND records with replacement roots.

FIGURE SPACE

Example: FA CON / SN cut projections

Highlight two output roots, same leaves, shared cell configuration.

OUTPUT: BIND / ROOTS / FANINS / REPLACE_ROOTS

Baseline: File-Order Candidate Loss

Details index GradMap integration

File-order greedy filtering ignores estimated area benefits of conflicting MO candidates.

41,476
Original MO candidate BIND records

7,024
FIFO retained (offline simulation)

16.9%
Overall retention rate

Design	Original MO	FIFO Retained	Dropped	Retention
EPFL hyp	25,440	4,180	21,260	16.4%
EPFL square	3,658	544	3,114	14.9%
EPFL log2	2,798	400	2,398	14.3%
EPFL multiplier	2,190	349	1,841	15.9%
EPFL voter	632	106	526	16.8%
ISCAS c6288	258	37	221	14.3%
ISCAS c7552	102	11	91	10.8%
EPFL adder	6	1	5	16.7%
All 364 designs	**41,476**	**7,024**	**34,452**	**16.9%**

Source: NF-direct full-corpus census, 364 benchmark entries; simulated parser-policy first-fit, not trained GradMap QoR.

Area-Aware MO Selection (MILP)

Maximize estimated area savings subject to non-overlapping MO candidates.

$$\begin{aligned} \max \quad & \sum_{i=1}^{N} \left(A_i^{SO} - A_i^{MO}\right) x_i \\ \text{s.t.} \quad & \sum_{i=1}^{N} a_{vi} x_i \le 1, \quad \forall v \in \mathcal{V} \\ & x_i \in \{0,1\}, \quad i = 1, \ldots, N \end{aligned}$$

FIGURE SPACE

Conflict: (A,B) vs (A,C)

Shared AIG node A $\Rightarrow$ at most one MO selected.

Implementation

Python + HiGHS (`highspy`); emit selected.matches.

Notation

x_i: select MO i; A_i^{SO}: SO witness area; A_i^{MO}: MO cell area;

$a_{vi} = 1$ iff AIG base node v belongs to MO candidate i; otherwise 0.

Objective weights estimate local SO-to-MO benefits; actual GradMap final mapped area is measured later.

Research Status & Next Steps

Details index Verification

Verified NF-direct discovery today; area-aware MILP is the next research step.

Completed / Verified

- Independent NF-direct MO discovery
- Boolean / structural legality
- SO witness + MOCAND integration
- 364-design candidate census

Next / Proposed

- Estimate SO witness area weights
- Solve MO selection with HiGHS
- Compare FIFO versus MILP
- Evaluate final mapped area / CEC

Research question

Can area-aware global pre-selection improve the MO opportunities available to GradMap?

QoR results intentionally left pending until the new code and benchmark runs are complete.

PROJECT 02 / ICCAD 2025 CAD CONTEST / PROBLEM C

Incremental Placement Optimization Beyond Detailed Placement

Simultaneous Gate Sizing, Buffering, and Cell Relocation

MY PROPOSED APPROACH

STA-Driven Incremental Timing ECO Search

Use OpenROAD / OpenSTA with policy-controlled ECO moves and global PPA evaluation.

Multi-Objective Timing ECO Optimization

02 / PROBLEM DEFINITION

Details index | STA targets | QoR / budget

Improve setup TNS toward zero while controlling total power, HPWL, and cell displacement.

OPTIMIZATION OBJECTIVES

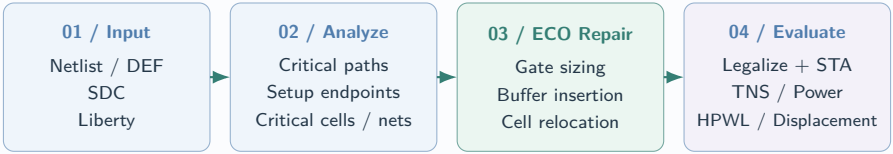
01 / Setup TNS
Reduce negative setup slack at violating endpoints.

02 / Total Power
Avoid excess power from ECO changes.

03 / HPWL
Control estimated wirelength after ECO.

04 / Displacement
Limit cell movement and keep placement legal.

BASIC ECO EVALUATION PIPELINE



Key challenge: a timing fix can worsen power or physical cost.

Five Automatic ECO Strategy Candidates

02 / CANDIDATE SET

Details index | STA targets | Endpoint coverage | ECO sequences

The candidate is an automatic repair policy and editable region, not a manually chosen cell replacement.

S1 / Global Setup

Whole design; automatic upsizing and VT selection.

S2 / Path-Restricted

Top-K path cells + nets; localized setup repair.

S3 / Overlap-Aware

Top-I shared cells, nets fixed; sizing / VT repair.

S4 / Local Buffering

Top-K nets, original cells fixed; buffer insertion and load splitting.

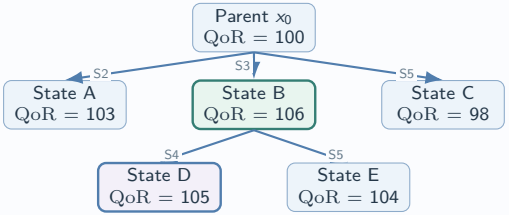
S5 / Power Recovery

Slack-rich cells / wider region; lower-power equivalent cells.

STA-Guided ECO Search: Best-First Branch List

No better child this round? Continue from the best unexplored state in the Branch List.

Search Tree Example



Bounded Branch List: D:105 > E:104 > A:103

Global Best: B (106) | **Next:** D (105)

A lower-scoring intermediate state may lead to a better ECO sequence later.

Heuristic Search Steps

01 / Pop Parent

Highest-QoR unexplored state from the **Branch List**.

02 / Generate + Evaluate

STA-guided ECO moves, legalization, and global QoR score.

03 / Update States

Save improved **Global Best**; queue legal child states.

04 / Next Branch

Expand best branch, **even with no new Global Best**.

QoR Results & Conclusion

02 / EARLIER ECO EVALUATION

Setup-focused timing repair and the resulting power / wirelength trade-off.

Setup Timing

16.7%
Median Setup TNS Improvement

14.6%
Average Setup WNS Improvement

Power & Wirelength

8%
Total Power Reduction

+2%
HPWL Increase

Conclusion

ECO operators can improve setup timing and power while accepting a small wirelength trade-off.

THANK YOU

Questions & Discussion

▸ Technical details

Technical Details

Back to project overview

QUESTIONS & DISCUSSION

Select a topic. Each detail page links back to the relevant summary slides.

Multi-Output Technology Mapping

D01 / SO baseline

D02 / Pair legality

D03 / SO witness

D04 / GradMap integration

D05 / Verification

STA-Driven Timing ECO

D06 / STA targets

D07 / Endpoint coverage

D08 / ECO sequences

D09 / QoR / budget

D10 / Parallel workers

ABC &nf: The Single-Output Baseline

Preserve original NF SO behavior while exposing its candidate search space.

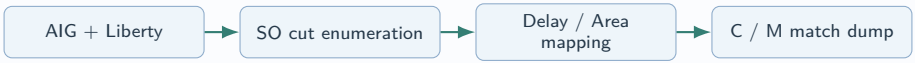


FIGURE SPACE

ABC NF: retained SO cuts vs selected mapping

One AIG root, several SO match choices, selected M highlighted.

Classic C Records

All retained single-output cut × library-cell matches.

Classic M Records

The final SO cover selected by ABC NF.

Invariant: enabling NF-direct MO discovery preserves classic C/M, vCellMapping, SO area and delay.

MO Pair Matching & Structural Legality

05 / CANDIDATE VALIDATION

Boolean matching is necessary, but not sufficient, for legal MO binding.

Boolean Matching

CON and SN roots match the cell's two truth-table projections.

Physical Compatibility

Same input support and a compatible pin permutation / polarity.

Structural Legality

Distinct roots; reject direct-dangling and TFI / cycle conflicts.

FIGURE SPACE

A valid FA/HA output pair in an AIG

Identify common leaves, roles CON/SN, and input phases.

Before MILP

Only individually legal, witness-backed MO records enter global selection.

SO Witness: Replacing Multiple SO Cells

A legal MO cell can replace a multi-level SO subcover with two outputs.

FIGURE SPACE

BEFORE: two SO cones → AFTER: one FA/HA

Show two output roots, common MO leaves and replaceable SO nodes.

SO Witness

Trace SO candidate matches back to the common MO input boundary.

REPLACE_ROOTS

Record base nodes participating in the candidate replacement.

External Demand

Shared internal roots may still need an SO implementation.

An SO witness is not necessarily the SO cover selected by ABC's final M records.

Connecting `&nf -Y` to GradMap

07 / DECISION INTEGRATION

Retain SO matches intact; expose selected MO candidates as SO/BIND decisions.

v3 MATCH CATALOG

```
C ...  % SO candidates
M ...  % selected SO cover

BIND <id>  FA/HA ...
ROOTS <CON> <SN>
FANINS <common leaves>
REPLACE_ROOTS ...

MOCAND <id> ...
```

GradMap MapBindingGroup

Each non-overlapping MO binding yields a decision: SO branch vs BIND branch.

FIGURE SPACE

GradMap SO/BIND decision graph

Optional screenshot of the decision groups and outputs.

MILP keeps opportunities; GradMap still decides whether to adopt each MO candidate.

Python + HiGHS: Verification Pipeline

Separate pre-selection quality from GradMap final mapped area.



Correctness Checks

Gate A: unchanged original NF mapping
Gate B: individual MO legality
Gate C: GradMap materialization + CEC

Experimental Comparison

SO-only vs FIFO vs MILP-selected MO; same AIG, library, GradMap settings, and seed.

FIGURE SPACE

Future Area / Runtime QoR Chart

Reserved until the MILP code and benchmark runs are complete.

Timing-Driven Target Extraction

Derive editable cells and nets from setup-violating paths and distinct endpoints.

```
report_checks -path_delay max
     -group_count 200
     -endpoint_count 3
     -slack_max 0
```

STA Report Scope

Setup paths may be PI→FF, FF→FF, FF→PO, or PI→PO, depending on constraints.

1 / Extract

Record endpoint, slack, path instances and nets.

2 / Deduplicate

Do not count the same violating endpoint repeatedly.

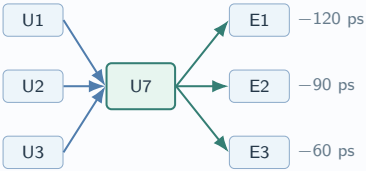
3 / Prioritize

Rank shared cells, critical nets, and candidate repair regions.

The STA report chooses where to search; OpenROAD performs the actual repair operation.

Slack-Weighted Endpoint Coverage

Prioritize shared cells by distinct endpoint coverage and violation severity.



One shared cell influences three distinct setup endpoints.

Distinct Endpoint Coverage

$Coverage(c) = |E(c)|$

Each endpoint is counted only once.

Slack-Weighted Priority

$$Priority(c) = \sum_{e \in E(c)} \max(0, -slack_e)$$

This priority guides search; the real ΔTNS still requires global STA after ECO.

Sequential ECO Strategy Combinations

Build a candidate by chaining compatible automatic repairs from a saved checkpoint.

S3 → S5

Repair shared timing-critical cells, then attempt power recovery if slack permits.

S4 → S3

Buffer selected critical nets, then re-target the remaining critical cells.

S3 → S4

Size critical cells first, then address residual net-delay bottlenecks.

Candidate Execution

Parent checkpoint

↓

Automatic repair pass 1

↓

Optional repair pass 2

↓

Legalize + re-evaluate global QoR

Operator order matters: buffering changes load; sizing changes driver strength.

Checkpoint Quality and Runtime Budget

Rank legal ECO checkpoints with consistent global QoR references, under one runtime budget.

ILLUSTRATIVE INTERNAL RANKING SCORE

$$\boxed{Q(x) = w_T\, g_T(x) + w_P\, g_P(x) - w_H\, p_H(x)}$$

$$g_T = \frac{|TNS_0| - |TNS_x|}{\max(|TNS_0|, \epsilon)}, \quad g_P = \frac{P_0 - P_x}{P_0}, \quad p_H = \frac{H_x - H_0}{H_0}$$

Quality-Based Search

Use the same baseline and weights when evaluating different branches; reject illegal or failed ECO states.

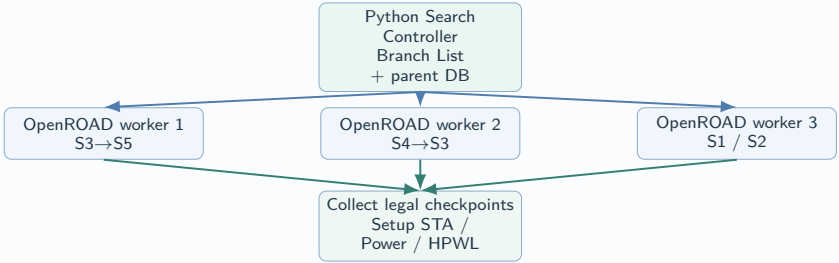
Runtime-Aware Execution

Batch processing consumes wall-clock time once; stop when the global execution budget is exhausted.

TNS: setup total negative slack; *P*: total power; *H*: HPWL. Weights define a search heuristic, not an official contest score.

Parallel Candidate Evaluation with OpenROAD

Independent workers run candidate ECO sequences from a selected Branch List checkpoint.



Worker Responsibilities

Apply ECO policy, legalize placement, update parasitics, collect full-design QoR, and save a restorable checkpoint.

Controller Responsibilities

Bound workers by available resources; update global best and Branch List using consistent QoR results.